1. Create a countdown timer that starts from 10 seconds and updates every second.

2. Display the remaining time in TextView and show “Time’s up!” when finished.

XML CODE:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity3">

    <TextView
        android:id="@+id/txtTimer"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text=""
        android:textSize="30dp"
        android:textStyle="bold"
        android:layout_marginBottom="30dp"
        />

    <TextView
        android:id="@+id/txtMsg"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:gravity="center"
        android:visibility="invisible"
        android:textSize="30dp"
        android:textStyle="bold"
        android:text="" />

</LinearLayout>
```

JAVA CODE:

```
package com.example.apppr15countdowmtimer;

import android.os.Bundle;
```

```
import android.os.CountDownTimer;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import java.util.concurrent.TimeUnit;

public class MainActivity3 extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main3);

        TextView txtTime = findViewById(R.id.txtTimer);
        TextView txtMsg = findViewById(R.id.txtMsg);

        CountDownTimer objTimer = new CountDownTimer(10 * 1000, 1000) {
            @Override
            public void onFinish() {
                txtMsg.setVisibility(TextView.VISIBLE);
                txtMsg.setText("Time's Up!");
            }

            @Override
            public void onTick(long millisUntilFinished) {

                long hour =
TimeUnit.MILLISECONDS.toHours(millisUntilFinished) % 24;
                long min
=TimeUnit.MILLISECONDS.toMinutes(millisUntilFinished) % 60;
                long sec =
TimeUnit.MILLISECONDS.toSeconds(millisUntilFinished) % 60;

                txtTime.setText(String.format("%02d : %02d : %02d", hour,
min, sec));

            }
        };

        objTimer.start();

    }
}
```

OUTPUT:

3:01
39%
00 : 00 : 06

3:05
39%
00 : 00 : 00
Time's Up!

3. Add a reset button to restart the timer from the beginning.

4. Modify the timer to accept user input for duration (e.g. from EditText).

5. Implement a timer that disables a button until countdown completes.

XML CODE:

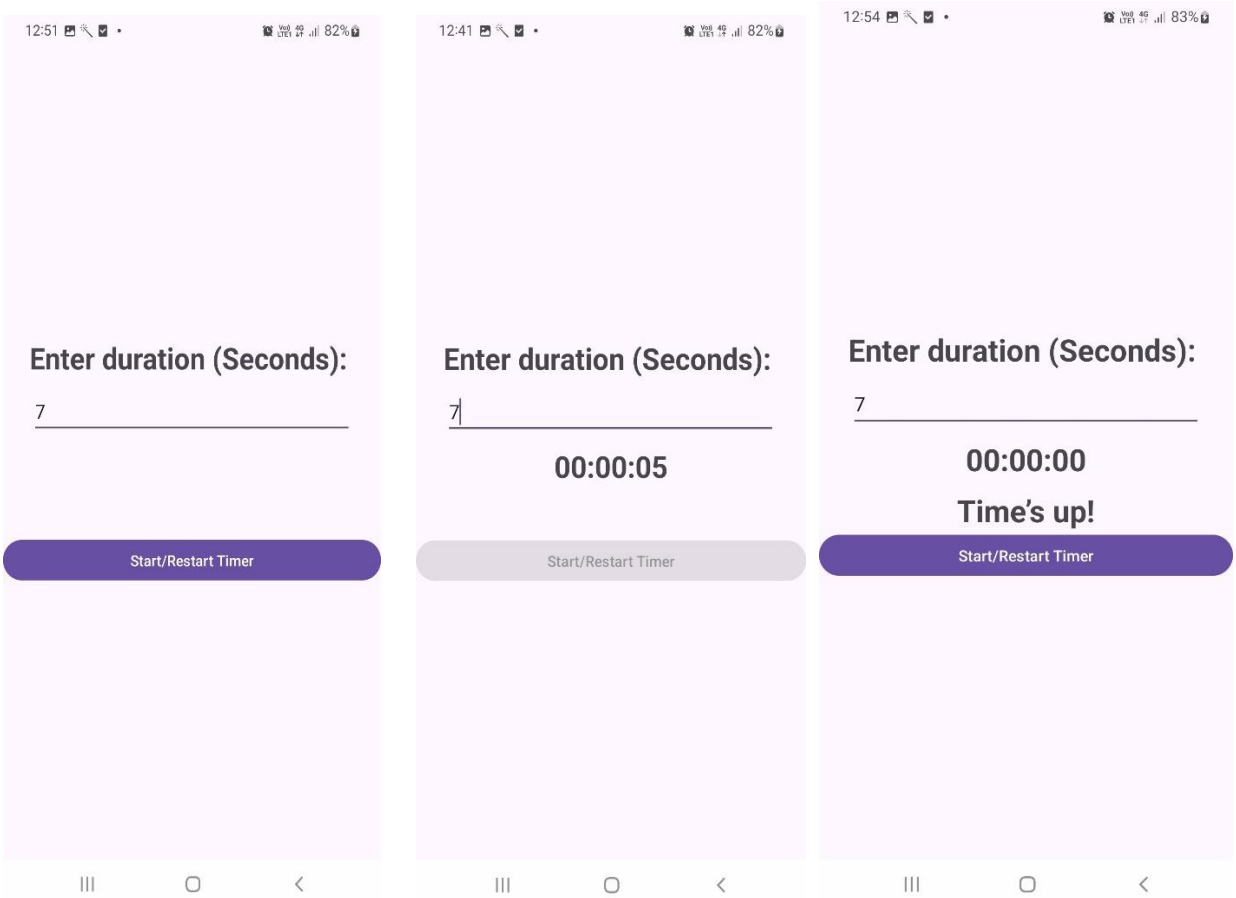
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:gravity="center"
        android:layout_marginBottom="10dp"
        android:text="Enter duration (Seconds): "
        android:textSize="30dp"
        android:textStyle="bold"
        />

    <EditText
        android:id="@+id/edtDuration"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:ems="15"
        android:layout_marginBottom="10dp"
        android:inputType="number"
        />


    <TextView
        android:id="@+id/txtTimer"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:gravity="center"
        android:text=""
        android:textSize="30dp"
        android:textStyle="bold"
      />
```

```
    <TextView
        android:id="@+id/txtMsg"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"
        android:gravity="center"
        android:text=""
        android:textSize="30dp"
        android:textStyle="bold"
        android:visibility="invisible"
        />

    <Button
        android:id="@+id/btnRestart"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Start/Restart Timer" />
</LinearLayout>
```

JAVA CODE:

```
package com.example.apppr15countdowmtimer;

import android.os.Bundle;
import android.os.CountDownTimer;
import android.os.Handler;
import android.os.Looper;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import java.text.DecimalFormat;
import java.text.NumberFormat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        TextView txt = findViewById(R.id.txtTimer);
        TextView txtMsg = findViewById(R.id.txtMsg);
        Button btnRestart = findViewById(R.id.btnRestart);
        EditText edtDuration = findViewById(R.id.edtDuration);


        btnRestart.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                txtMsg.setVisibility(View.INVISIBLE);

                if (edtDuration.getText().toString().isEmpty()) {
                    Toast.makeText(MainActivity.this, "Enter Duration",
Toast.LENGTH_SHORT).show();
                    return;
                }


                long lDuration =
Long.parseLong(edtDuration.getText().toString());

                CountDownTimer objCountDownTimer = new
CountDownTimer(lDuration * 1000, 1000) {
                    @Override
                    public void onFinish() {
                        btnRestart.setEnabled(true);
                        txtMsg.setVisibility(View.VISIBLE);
                        txtMsg.setText("Time’s up!");
                    }

                    @Override
                    public void onTick(long millisUntilFinished) {
                        btnRestart.setEnabled(false);

                        long hour = (millisUntilFinished / 3600000) % 24;
                        long min = (millisUntilFinished / 60000) % 60;
                        long sec = (millisUntilFinished / 1000) % 60;

                        Handler hd = new Handler(Looper.getMainLooper());
                        hd.post(new Runnable() {
                            @Override
                            public void run() {
                                NumberFormat f = new DecimalFormat("00");
                                txt.setText(f.format(hour) + ":" +
f.format(min) + ":" + f.format(sec));
                            }
```

```
                        });
                    }
                };

                objCountDownTimer.start();
            }
        });
    }
}
```

OUTPUT: